ANALYSIS OF NETWORK PERFORMANCE PARAMETERS IN A 5G STANDALONE SYSTEM

INTRODUCTION:

This mini-project focuses on the analysis of key network performance parameters in a 5G standalone system, namely uplink, downlink, latency, throughput, jitter, and propagation delay, which collectively determine the quality and efficiency of data communication. Uplink and downlink characterize the directional flow of data between user equipment and the 5G core network, while latency and propagation delay influence end-to-end responsiveness. Throughput reflects the effective data transfer capability of the network, and jitter represents the variation in packet delay, significantly impacting real-time and delay-sensitive applications.

By studying these parameters in a 5G SA environment, the project aims to gain insights into network behaviour, performance characteristics, and potential limitations under practical operating conditions. The analysis contributes to a deeper understanding of how 5G standalone networks meet the stringent performance requirements of next-generation applications and provides a foundation for future optimization and research in advanced wireless communication systems.

TOOLS USED:

To evaluate the performance of the 5G Standalone (SA) network, a practical test setup was implemented using a 5G-enabled mobile device as the User Equipment (UE) in which the 5G registered SIM card inserted and a gNodeB (gNB) connected to the 5G core network.

1. 5G-Enabled Mobile Device (User Equipment)

A mobile phone with 5G support was configured to act as the User Equipment (UE) in the 5G Standalone network. The UE was connected to the gNodeB and used to initiate performance measurements. To analyse both uplink and downlink performance, the UE and the network node were configured to operate alternately as client and server, depending on the direction of traffic under study.

2. Ping Utility (installed on UE)

The ping tool was used to measure end-to-end round-trip latency between the UE and the gNodeB or core network server using ICMP echo request and reply messages. Ping was executed from the UE when it acted as the client to assess network responsiveness. By interchanging the source and destination roles, latency behaviour in both uplink and downlink directions could be observed.

3. iPerf Tool (installed on UE)

The iPerf tool was employed to generate traffic for measuring throughput, latency, jitter, and packet loss. iPerf was configured such that either the UE or the network node acted as the server, while the other acted as the client, allowing separate evaluation of uplink and downlink performance. UDP mode was primarily used to analyse delay-sensitive metrics such as jitter and packet delay variation, while TCP mode was used to assess achievable throughput.

4. Terminal Emulator – Terminator (installed on SA network)

A terminal emulator application, Terminator, was installed on the SA network to provide command-line access for executing network performance tools. The terminal will be enabled to interact directly with the gNodeB by logging into the system using the IP address of the gNodeB.

PARAMETERS STUDIED AND ITS SIGNIFICANCE:

1)Uplink Data-rate

Uplink refers to the data transmission path from the user equipment (UE) or end device (such as a mobile phone, IoT node, or computer) to the network infrastructure, typically a base station, access point, or server.

Significance

- Determines how efficiently data such as sensor readings, voice input, video uploads, and control signals are sent from the user to the network.
- Critical for applications that rely on real-time data transmission, such as video conferencing, cloud computing, telemetry, and uplink-heavy IoT systems.
- Limited uplink capacity can cause packet delays, data loss, and degraded application performance, especially in wireless networks where power and bandwidth constraints exist.

2)Downlink Data-rate

Downlink is the data transmission path from the network infrastructure to the user equipment. It carries information such as web content, video streams, control responses, and application data.

Significance

- Directly affects user experience, as most applications are downlink-dominant (e.g., web browsing, video streaming, software downloads).
- Higher downlink bandwidth enables faster content delivery, reduced buffering, and improved quality of service (QoS).

3)Jitter

Jitter is the variation in packet delay during data transmission over a network. It represents the inconsistency in the arrival time of consecutive packets compared to their expected arrival intervals.

Significance

- High jitter negatively impacts real-time applications such as Voice over IP (VoIP), video conferencing, online gaming, and live streaming.
- Causes audio distortion, video frame drops, and synchronization issues even when bandwidth is sufficient.
- Jitter analysis helps assess network stability and congestion levels, making it an important parameter in performance monitoring and optimization.

4)Latency

Latency is the total time taken for a data packet to travel from the source to the destination across a network. It includes transmission delay, propagation delay, processing delay, and queuing delay, and is typically measured in milliseconds (ms).

Significance

- Low latency is critical for real-time and interactive applications such as voice calls, video conferencing, online gaming, and remote-control systems.
- High latency leads to delayed responses, reduced interactivity, and poor user experience, even if bandwidth is high.
- Latency measurement helps evaluate network responsiveness and the effectiveness of routing, scheduling, and resource allocation mechanisms.

5)Throughput

Throughput is the actual rate of successful data delivery over a network, representing the amount of data received at the destination per unit time. It is typically measured in bits per second (bps) and reflects the effective utilization of available bandwidth.

Significance

- High throughput enables faster data transfers, smoother video streaming, and efficient application performance.
- Throughput is affected by factors such as network congestion, protocol overhead, packet loss, and channel conditions.
- Throughput analysis helps assess network capacity, efficiency, and performance under load, making it a key metric in network evaluation and optimization.

COMMANDS AND RESULTS:

For testing Data rate:

1) Downlink Test:

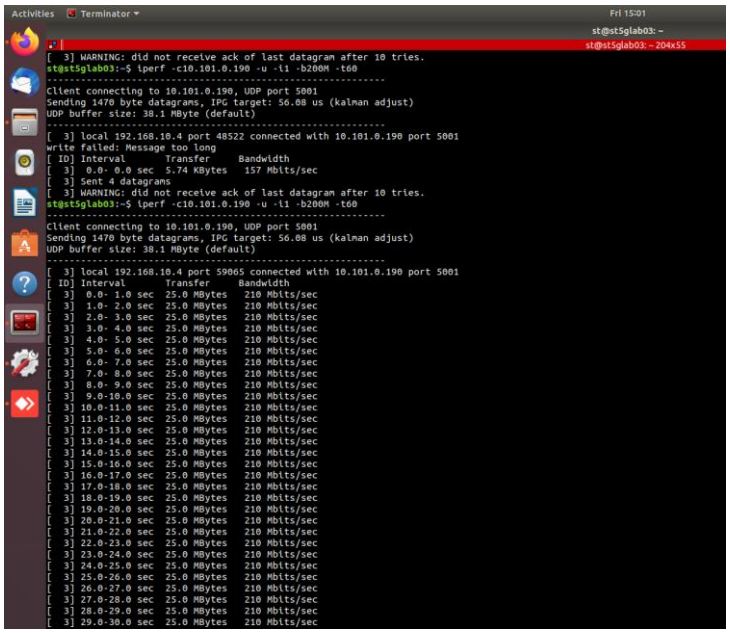
On the UE side:

- Open the iperf application in UE (mobile phone).
- Run iperf –s –u –i1
- Iperf is already enabled in the mobile application.
- So, we should type only –s –u –i1
- Iperf - Launches the iperf network performance measurement tool.
- -s (Server mode) - Runs iperf as a server. The server listens for incoming traffic. It does NOT send data, it only receives. The device running -s is the receiver.
- -u (UDP mode) - Uses UDP protocol instead of TCP. Required to measure: Jitter & Packet loss. UDP does not retransmit packets, making it suitable for real-time traffic analysis.
- -i 1 (Interval reporting) - Displays performance statistics every 1 second.

On the NGC side:

- Open the terminator.
- Login into the gNodeB by entering its IP address and password.
- Run iperf -c<DESTINATION_IP> -u -i1 -b200M -t60
- Iperf- Starts the iperf traffic generation tool.
- In iperf, Client = sender and Server = receiver
- -c <DESTINATION_IP> (Client mode) - Runs iperf as a client. The client sends data to the specified destination IP. The destination must be running iperf in server mode (-s).
- -u (UDP mode) - Uses UDP protocol.
- -i 1 (Interval reporting) - Displays transmission statistics every 1 second. Helps observe performance variation over time.
- -b 200M (Bandwidth / Data Rate) - Sets the offered data rate to 200 Mbps. This is the intended transmission rate, not the guaranteed throughput.
- -t 60 (Test duration) - Runs the traffic generation for 60 seconds. Provides stable and averaged performance results.

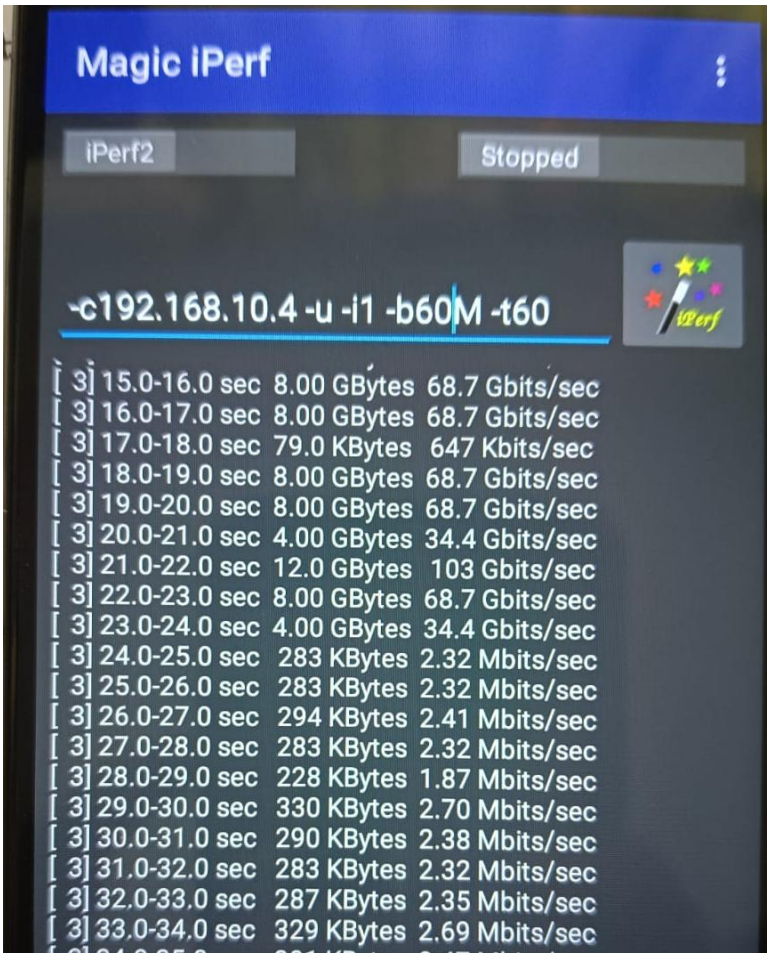
  The maximum downlink data rate observed in the Signaltron-based 5G SA setup was greater than 340 Mbps. The offered data rate can be configured by the user using the -b parameter in the iperf command.



2) **Uplink Test:**

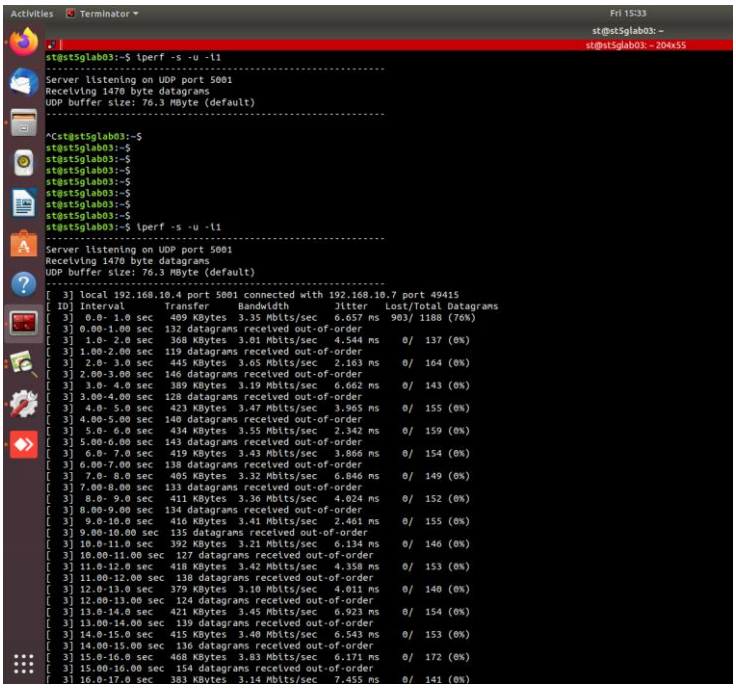
On the UE side:

- Open the iperf application in UE,
- Open the iperf application in UE (mobile phone).
- Run iperf –c [UE ip] –u –i1 –b200M –t60
- Iperf is already enabled in the mobile application.
- So, we should type only –c [UE ip] –u –i1 –b200M –t60
- This command runs iperf in UDP client mode to send traffic to the UE at a configured data rate of 200 Mbps for 60 seconds, while reporting performance statistics every 1 second.
- Here the UE is the gNodeB and its IP address is 192.168.10.4



On the NGC side:

- Open the terminator. Login into the gNodeB by entering its IP address and password.
- Run iperf –s –u –i1
- This command starts iperf in UDP server mode, where the system receives UDP traffic and reports performance statistics every 1 second.



- During the uplink test, the data rate was configured as 60 Mbps at the UE using the iperf bandwidth option. However, the achieved uplink throughput observed at the receiver varied between 3 and 4 Mbps. This indicates that while a higher data rate was offered by the sender, the actual throughput was constrained by uplink resource allocation, UE transmit power limitations, and network scheduling mechanisms in the 5G Standalone setup.

3) Latency Measurement

Option 1: Latency Measurement Using ping (ICMP-based)

Why this method is used

- ping measures round-trip time (RTT) between the UE and the NGC.
- It operates at the network layer (ICMP).
- It is:
    - Simple
    - Widely supported
    - Independent of application traffic

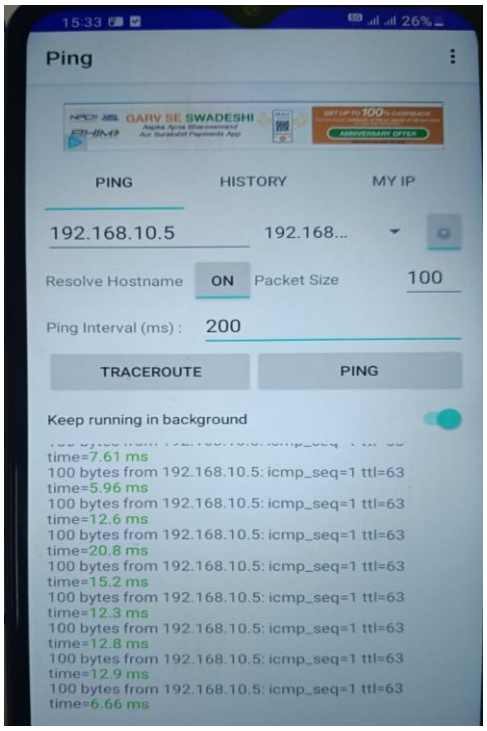
It measures the RTT (Round Trip Time).

Advantages

- Supported in all Linux versions
- Stable and consistent
- Suitable for baseline latency measurement

Limitation

- Does not represent application-level traffic
- Measures control-plane ICMP packets, not user data packets



Latency Test Configuration

Latency was measured using ICMP echo requests (ping) from the UE to the 5G core network gateway. The test configuration parameters are as follows:

Parameter	Value
Destination IP	192.168.10.5
Protocol	ICMP
Packet size	100 bytes
Ping interval	200 ms
Number of packets	*Continuous (sampled values recorded)**
Resolve hostname	Enabled

** Due to the continuous ping function of the mobile app, sampled RTT values were recorded and analyzed.*

Latency was measured using ICMP echo requests (ping) from the UE to the 5G core network gateway.

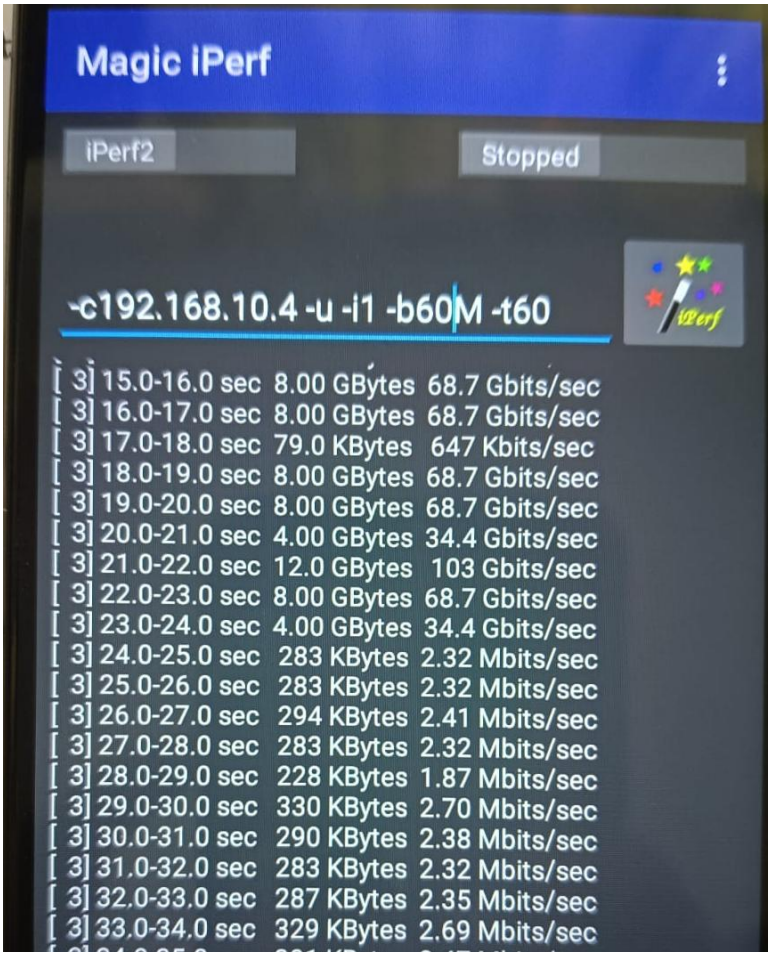
- Latency was measured using ICMP echo requests (ping) from the User Equipment (UE) to the core network gateway. The test was conducted with a packet size of 100 bytes and a ping interval of 200 ms. The observed round-trip latency values ranged between 5 ms and 21 ms, with an average latency of approximately 10–12 ms. These results indicate low and stable end-to-end delay in the 5G Standalone network. The measured latency values fall within the expected performance range for a 5G SA setup, confirming efficient packet forwarding between the UE and the core network.

Option 2: Latency Measurement Using iperf (UDP-based)

Why this method is used

- iperf operates at the application layer
- Uses UDP traffic, similar to real-time 5G services
- Intended to measure:
    - Throughput
    - Jitter
    - Packet loss
    - Latency (only in newer versions)

- The throughput, jitter, and packet loss were measured for the uplink direction.
- Accordingly, the UE was configured as the iperf client which sends the data by running the command
- Run iperf -c [gNodeB IP] -u -b60M -t60
- This command runs iperf in UDP client mode to transmit traffic toward the gNodeB at a configured data rate of 60 Mbps for a duration of 60 seconds.



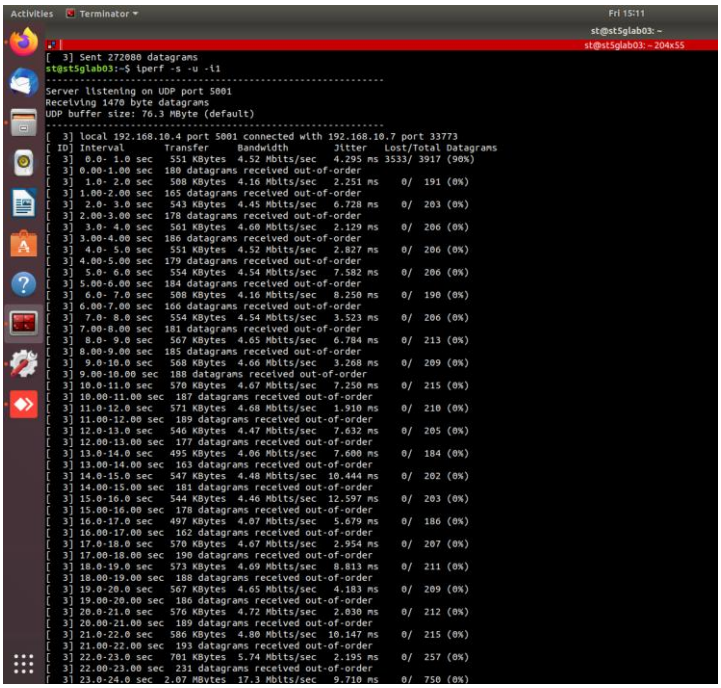
On the NGC side:

- Open the terminator. Login into the gNodeB by entering its IP address and password.
- Run iperf -s -u -i1

Explaining iperf UDP Server Output Metrics

Parameter	What it actually means
Interval	Time window over which measurements are calculated (e.g. 1 second)
Transfer	Total data successfully received during that interval
Bandwidth	**Throughput** – successful data rate at the receiver
Jitter	Variation in packet arrival time (delay variation)
Lost/Total Datagrams	Number of packets lost vs sent (reliability metric

Output:



- The iperf server output shows the achieved throughput, jitter, and packet loss for the tested link. The bandwidth value represents the actual throughput, indicating the rate of successfully received data at the receiver. The observed throughput was lower than the configured data rate due to network scheduling and resource constraints. The measured jitter values were within acceptable limits, and packet loss was minimal, indicating stable and reliable data transmission.